

State-Dependent Climate Alpha: Green-Minus-Brown Industry Timing

MFE 230G – Active Asset Management – Final Project

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Executive Summary

We test whether a factor-neutral Green-minus-Brown industry spread (the five lowest-emissions Fama-French industries minus the five highest) earns money specifically when climate-transition attention is unusually high, and whether that depends on rates, oil, inflation, or the business cycle. Our conclusion is do not implement. The only period where the strategy actually works is COVID, 2020-2021. Once we test it on data from after we built it, and once we account for the number of things we tested, the result does not hold up. Every other window we checked, including the most recent 12 and 18 months, is small, not statistically significant, or losing money after costs. We also tried a smoother version of the signal. It loses less money and swings around less, which is a real improvement, but it still does not show a reliable edge. The full case for this, including where our AI assistant helped and where we had to correct it, is below.

1 What Did You Try?

1.1 Thesis

Our thesis, stated so it can be proven wrong: the Green-minus-Brown spread is not a stable, always-on source of return, but its factor-neutral part becomes profitable to trade – short Brown – specifically in the months right after climate attention spikes above its own historical range. If a holdout period we never touched while building the strategy shows no such profit, the thesis is wrong. We ran that exact test, and it says the thesis is wrong (Section 2).

1.2 Data

We used four monthly data files. None of them have missing values that matter for this analysis.

- `ff49_industry_monthly.csv`: returns for the 49 Fama-French industry portfolios, from July 1926 to July 2026, as decimals.
- `ff3_factors_monthly.csv`: the market, size, and value factors plus the risk-free rate, same dates.
- `emissions_ff_industry.csv`: a single snapshot of emissions intensity by industry. It only covers 41 of the 49 industries. Eight are missing, including Oil, so they are simply left out of the ranking.
- `macro_monthly.csv`: climate attention, the 10-year Treasury yield, WTI oil, CPI, CFNAI, the NBER recession flag, and GSCPI. Each series starts at a different date (attention in 1985, oil in 1986), and the file itself starts in 1854.

We checked that the dates line up correctly across all four files. The notebook only uses relative paths (`data/*.csv`), so it runs as-is from a fresh copy of this folder.

1.3 Implementation

Building Green minus Brown. We rank industries by emissions intensity, take the five lowest as Green (Fun, REst, Drugs, Telcm, Fin) and the five highest as Brown (Util, Ships, Aero, Steel, BldMt). This ranking comes from the data, not from us picking the industries by hand, and it matches what we expected before running anything.

$$GB_t = R_t^{Green} - R_t^{Brown}$$

Removing factor exposure. We run a rolling 60-month regression of the spread on the three Fama-French factors and use the resulting betas and intercept one month later than they were estimated, so nothing in month t 's hedge could have used month t information. This gives us three different things, and it matters that we keep them separate: the raw spread GB_t ; the hedged return $GB_t - \hat{\beta}_{t-1}'F_t$, which still contains whatever alpha the spread had; and the residual ε_t , which also removes that alpha and is the thing every signal below is actually trying to predict.

Building the signals. Climate attention is $z_{60}(\log(1 + \text{attention}))$, a rolling z-score of log attention. We build the same kind of z-score for the change in the 10-year rate and for oil's log return. We also use last month's inflation level, last month's change in inflation, last month's CFNAI reading, the recession flag, and last month's GSCPI. All the rolling windows are 60 months long and need at least 36 observations before they turn on.

Cleaning the attention signal. We try to strip macro noise out of attention by predicting it each month with a ridge regression trained only on the past 120 months (minimum 60 usable months, ridge penalty 0.10), then taking the leftover as the "clean" signal: $A_t^{pure} = A_t - \hat{E}_{t-1}[A_t | X_t]$. The predictors here are only ones available in real time: rate shocks, oil shocks, and lagged inflation and activity. We also tried a second version that adds the recession flag and GSCPI, but only as a check afterward, because neither one is actually available in real time (the recession flag is dated 6-18 months late, and GSCPI gets revised).

The trading rule. We flag a month as "extreme" when attention is above the 80th percentile of everything seen so far – using only past data, so nothing here looks ahead. Whenever attention crosses up through that line, we start a Short-Brown position and hold it for either 3 or 6 months.

Sizing and costs. The position size targets 5% annualized volatility based on the trailing 36 months, capped at 1.0, and is lagged one more month so a position decided using information through month t only earns the month $t + 1$ return. We also carry a factor overlay to keep the position hedged, and we track and charge turnover on both the position and the overlay: 10bp on the asset leg, 5bp on the market-factor overlay, and 25bp on the size and value overlay, per unit of monthly turnover.

How we test it. For every return series we report: annualized net return, annualized volatility, Sharpe ratio, max drawdown, annualized turnover, factor alpha with a Newey-West t-stat (6 lags, and we checked 12 and 18 lags too), how many months the strategy was actually active, the Information Coefficient between the signal and the return it's trying to predict, a 12-month block bootstrap with 5,000 draws, and a Holm-Bonferroni correction across everything we tested. We look at the full sample (2010-2026), a validation window (2010 through mid-2022), a holdout we never touched until the design was frozen (mid-2022 through mid-2026), pre-COVID, COVID, the 2022-2024 inflation/rate period, and the most recent 12 and 18 months.

2 What Did You Learn?

2.1 Is this actually alpha, or a repackaged factor bet?

We checked this by regressing the traded strategy’s returns, not the raw spread, on the market, size, and value factors. The raw spread does load heavily on value: -0.24 over the full sample, -0.35 since 2010. That is a real growth-versus-value bet hiding inside what looks like a climate trade. After we hedge it, the size loading disappears completely in every version we tested ($|t| < 1.2$), and the value loading drops a lot, down to about -0.05 to -0.06 . But it does not disappear. It is still statistically significant, with t-stats around -2 to -2.4 , in every version. So the honest answer is in between: the hedge removes most of the factor exposure, not all of it, and a small piece of the return we report is still a value bet rather than a climate bet.

2.2 Is performance stable, or driven by one regime?

It is driven by one regime. Table 1 and Figure 1 both show the same thing. COVID, on its own, produces a large and significant gain: the 3-month rule earns 5.94% net a year with an alpha of 6.03% ($t = 3.21$), and the 6-month rule earns 7.41% net with an alpha of 5.23% ($t = 2.07$). Every other window looks weak or negative. The holdout, which runs from August 2022 to July 2026 and was never touched while we designed the strategy, loses money in every version we tested, between 2.1% and 3.8% a year, with t-stats around -1.7 to -2.2 . The two most recent windows tell the same story: over the last 12 months every version lost between 1% and 9% a year (t as low as -2.7), and over the last 18 months every version lost between 3% and 8% (t as low as -4.7). We are reporting this even though it looks bad, because that is the point of a holdout.

Table 1: Net performance of the discrete Short-Brown strategies

Sample	Strategy	Net return	Sharpe	FF3 alpha	$t(\alpha)$
2010–Jul. 2022	Original Brown 3m	1.31%	0.41	1.33%	1.47
	Pure Brown 3m	1.41%	0.44	1.44%	1.50
	Original Brown 6m	2.45%	0.59	2.25%	2.23
	Pure Brown 6m	2.59%	0.60	2.51%	2.46
Aug. 2022–Jul. 2026	Original Brown 3m	-3.82%	-1.15	-4.03%	-2.19
	Pure Brown 3m	-2.17%	-0.93	-2.21%	-1.93
	Original Brown 6m	-2.14%	-0.49	-2.28%	-1.66
	Pure Brown 6m	-3.04%	-0.74	-2.71%	-1.94

Returns and alphas are annualized and net of stated transaction-cost assumptions. HAC t -statistics use six lags. “Pure” attention is orthogonalized with past-only macro coefficients.

2.3 Turnover, costs, and position sizing

Annualized turnover runs from about 2.7x to 5.2x a year. That comes from the monthly overlay rebalancing plus the threshold entries and exits, not from us choosing to trade a lot. It costs 0.3 to 0.6 percentage points a year. That is real money, but it is not why the strategy fails: the gross returns are already small, 0.6% to 1.7% a year over the full sample, so costs eat a big share of a thin edge without being the main reason the holdout loses money – the holdout losses are much bigger than the cost drag on their own. On sizing: we target volatility directly instead of using the HW02 rule $h^* = \alpha_n / (2\lambda\omega_n^2)$. The two line up when $\lambda_t = \alpha_t / (2\sigma_{tgt}\omega_t)$, and if you back out what λ that implies here, it runs from about 0.17 for the

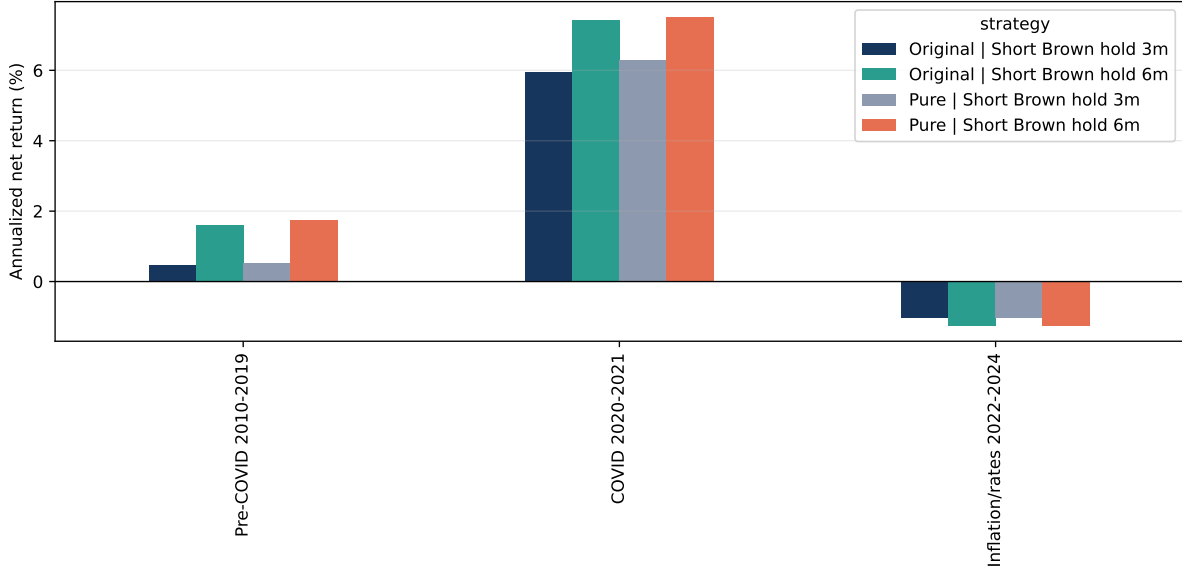


Figure 1: Net performance changes sign across economic regimes.

3-month raw version up to about 2.2 for the 6-month purified version. We only report this as a diagnostic. The strategy itself never uses α_t to decide how big a position to take.

2.4 Signal quality: the Information Coefficient

The IC, the correlation between the signal and the return it is trying to predict, has the right sign in validation but is weak: about -0.08 to -0.09 for raw and purified attention, and a stronger, individually significant -0.18 to -0.20 for the smoothed version in Section 2.5. In the holdout, it flips sign for every version we tested, to about $+0.12$ to $+0.15$. That matters on its own, separate from the strategy returns: it is not just that the trading rule lost money out of sample, it is that the relationship the whole idea depends on ran backwards.

2.5 An improvement: continuous conditioning (does it change the conclusion?)

Instead of an on/off threshold and a fixed 3- or 6-month hold, we tried a smoother version: a weight that rises gradually as attention gets more extreme, and fades gradually afterward (floor $c = 0.5$, half-life $H = 3$ months – chosen to match the spirit of the original design, not tuned to make the results look good). It cuts full-sample volatility by 55-65% and max drawdown by 60-75%, and it loses 75-85% less than the discrete version in the holdout. A paired bootstrap favors it over every discrete version we compared it to (raw p-values of 0.01 to 0.07), but none of those comparisons survive a Holm-Bonferroni correction, and a placebo test where we shuffle the calendar does not clear the usual significance bar either ($P(\text{placebo} \geq \text{real}) \approx 0.12$). So this does not change our conclusion. It is a better way to size a signal we still cannot confirm, not a better signal.

2.6 Robustness

The near-zero unconditional result holds up if we widen each leg from 5 to 10 industries, and it holds up if we drop Utilities, the single highest-emissions industry, out of Brown entirely. We also looked at every

strategy and period combination we tested anywhere in this project: 8 strategy versions across 4 periods, 32 combinations in total. After a Holm-Bonferroni correction, only two of those 32 survive, and both are in the COVID window: the discrete 3-month rule ($t = 3.21$) and the smoothed version ($t = 3.28$). Several other results looked convincing on their own. None of them hold up once we account for how many things we tested.

2.7 Critical evaluation of the AI’s role

We used two GenAI tools at two different stages. Before writing any code, we used ChatGPT to inspect the raw notebook and data files and produce a replication blueprint – a plan, not code. It locked in exact parameters (window lengths, thresholds, cost assumptions) before any code existed, so the design did not drift later. Once the plan was set, we used a GenAI coding assistant, Claude, through Claude Code, for idea generation, implementation, and robustness testing. It was most useful for building statistical machinery correctly and quickly – rolling regressions with proper standard errors, the block bootstrap, the multiple-testing correction – and for turning a vague ”test an improvement” instruction into one specific, falsifiable design instead of several vague ones. Both stages needed correcting at least once. The more serious case: an early version compared the macro R-squared of the raw and purified signals using two different sample windows, which would have made the cleanup look more effective than it actually was. We caught it, and the corrected numbers are the ones in this report.

References

- [1] Bolton, P., and M. Kacperczyk (2021). Do Investors Care about Carbon Risk? *Journal of Financial Economics* 142, 517–549.
- [2] Pástor, L., R. Stambaugh, and L. Taylor (2021). Sustainable Investing in Equilibrium. *Journal of Financial Economics* 142, 550–571.
- [3] Pástor, L., R. Stambaugh, and L. Taylor (2022). Dissecting Green Returns. *Journal of Financial Economics* 146, 403–424.
- [4] Aswani, J., A. Raghunandan, and S. Rajgopal (2024). Are Carbon Emissions Associated with Stock Returns? *Review of Finance* 28, 75–106.
- [5] Zhang, S. (2025). Carbon Returns across the Globe. *Journal of Finance*. DOI: 10.1111/jofi.13402.
- [6] Eskildsen, M., M. Ibert, T. Jensen, and L. Pedersen (2026). Realized Returns to Green Investing: A Global Replication. SSRN 6527562.
- [7] Gârleanu, N., and L. Pedersen (2026). Climate Risk Pricing. SSRN 5978854.
- [8] Călin, A., R. Lupu, and R. Topa (2026). Climate Transition Risk Premium in Equity Markets. *Borsa Istanbul Review*, 100895.